Technical Characterization and Systematic Failure Modes in Historical Oil Paint Films: A Study of Pigment-Medium Interactions and Degradation Mechanisms

The historical evolution of oil painting materials represents a complex intersection of empirical artistry and unwitting chemical experimentation. A painting is not a static image but a dynamic chemical system where the constituent layers—support, ground, paint, and glaze—continuously interact with each other and their environment over centuries. The stability of these works depends heavily on the chemical compatibility of pigments and binders, a relationship that has frequently resulted in unforeseen degradation phenomena. Investigations by major conservation institutions, including the National Gallery, London, the Metropolitan Museum of Art, and the Getty Conservation Institute, have identified several critical failure modes, including the leaching of potash from smalt, the formation of heavy-metal soaps in lead- and zinc-containing paints, and the structural complexities of red lake substrates.

The scientific analysis of these materials has shifted the focus of art history from purely stylistic concerns to a deep understanding of materiality. By employing advanced analytical techniques such as Gas Chromatography-Mass Spectrometry (GC-MS), Fourier Transform Infrared Spectroscopy (FTIR), and Synchrotron-based X-ray methods, researchers have begun to decode the "inherent vice" of certain traditional palettes. This report synthesizes technical findings from major technical bulletins and research archives to provide a comprehensive overview of how pigment-medium interactions dictate the longevity and appearance of European oil paintings.

Chemical Instability and Color Loss in Smalt-Based Paint Layers

Smalt, a blue cobalt-containing potash glass pigment, was widely adopted in European oil painting from the late fifteenth century through the seventeenth century.[1] Its popularity stemmed largely from its affordability compared to natural ultramarine (lapis lazuli), yet its chemical instability has led to profound alterations in many masterworks, notably those by Rembrandt, Paolo Fiammingo, and various Northern European masters.[2]

Mechanisms of Potash Leaching and Medium Alteration

The degradation of smalt is primarily characterized by the loss of its vibrant blue hue and the simultaneous yellowing of the surrounding oil medium.[1] Research at the National Gallery indicates that this process is driven by the leaching of potassium ions (K^+) from the glass matrix into the binding medium.[3] As the alkali component is lost, the coordination environment of the cobalt ion (Co^{2+}) within the glass shifts. In a stable glass matrix, the Co^{2+} ion is typically in a tetrahedral coordination, which provides the characteristic deep blue color; as the potassium is leached and the glass becomes hydrated, the coordination can shift toward an octahedral environment, resulting in a pale pinkish or greyish tone that, when combined with the yellowed oil, appears brownish-yellow or greenish-grey.[1]

Technical analysis using Scanning Electron Microscopy with Energy Dispersive X-ray analysis (SEM-EDX) has quantified this loss with precision. In well-preserved samples, smalt may contain approximately 13-15% weight of potassium expressed as K_2C.[1] Conversely, in degraded layers, the potassium content often drops to as low as 1-2%.[3] This migration of potassium ions into the oil medium is not a neutral event; it triggers secondary reactions that compromise the binder's structural integrity. FTIR microscopy has confirmed that the leached potassium reacts with free fatty acids in the oil—released through the hydrolysis of triglycerides—to form potassium soaps.[3] These soaps can accumulate within the paint film or manifest as surface crusts.

The high surface area of crushed smalt glass, combined with its immersion in an acidic drying oil environment, facilitates this leaching process. Historical recipes often involved grinding the glass to a coarse powder to maintain color, but even these larger particles are susceptible.[1] Furthermore, the acidic nature of the drying oil itself can act as a leaching agent, pulling the alkali components out of the glass silicate structure.

Physical Manifestations and Surface Defects

Beyond the loss of color, smalt-containing paint often exhibits physical defects such as drying cracks and patchy blanching.[1] Blanching, where the paint appears lighter at the surface, is frequently attributed to the disintegration of the binding medium or the formation of efflorescence, which increases light scattering.[4] The presence of calcium oxalate in surface crusts, similar to those found on degraded vessel glass, suggests that atmospheric factors and moisture play a significant role in the deterioration of smalt-pigmented layers.[1] These crusts may also contain carbonates and sulfates, analogous to the weathering products found on historical window glass.[1]

Component Feature	Status in Preserved Smalt	Status in Degraded Smalt	Analytical Evidence

Visual Color	Deep vibrant blue	Greyish-yellow / Brownish	Visible light microscopy [2]
Potassium Content (K_2C)	13% - 15% wt	1% - 2% wt	SEM-EDX [3]
Medium Condition	Clear, polymerized oil	Yellowed, disintegrated	GC-MS and FTIR [4]
By-products	None	Potassium soaps, oxalates	FTIR microscopy [3]
Optical Effect	Transparent glaze	Light-scattering blanching	Cross-section analysis [1]

The presence of lead white (basic lead carbonate, $2PbCO_3 \cdot Pb(OH)_2$) in mixture with smalt has been observed to offer a stabilizing effect in some instances. Cross-sections from Paolo Fiammingo's *The Sons of Boreas pursuing the Harpies* (National Gallery, NG 5467) show that in upper layers where smalt was mixed with significant lead white, the pigment remained well-preserved; in contrast, in lower layers with minimal lead white, the smalt had almost entirely lost its color and was severely depleted of potassium.[3] This suggests that lead white might buffer the acidic environment or provide reactive sites for fatty acids that would otherwise target the smalt glass.

Heavy-Metal Soap Formation: Lead and Zinc Carboxylates

Perhaps the most pervasive threat to the structural integrity of oil paintings is the formation of heavy-metal soaps. These are salts formed by the reaction between metal cations—typically lead (Pb^{2+}), zinc (Zn^{2+}), or copper (Cu^{2+})—and free fatty acids produced by the hydrolysis of the oil binder.[5] While lead soaps are common in Old Master paintings due to the ubiquity of lead white and lead-tin yellow, zinc soaps are a critical concern for nineteenth- and twentieth-century works.[8]

Lead Soap Inclusions and Protrusion Mechanisms

Lead soaps (primarily lead palmitate and lead stearate) often aggregate into translucent white lumps or "inclusions" within the paint or ground layers.[5] These aggregates can grow to diameters of 100–200 μm, significantly larger than the original pigment particles.[5] As they

expand, they exert internal pressure on the paint film, eventually erupting through the surface to create a granular, gritty texture or "craters".[5] This phenomenon has been documented in hundreds of paintings dating from the fifteenth to the twentieth centuries.[5]

Historically, these inclusions were variously interpreted as gas bubbles, "bubbles" from aqueous media like egg tempera, or even insect eggs.[11] However, modern analytical techniques like FTIR, Py-GC-MS, and Solid-State Nuclear Magnetic Resonance (ssNMR) have conclusively identified them as lead carboxylates.[5] The formation process involves the transport of free fatty acids through the paint matrix to the lead-containing pigment sites. Once the lead ion is released from the pigment (e.g., from lead white or lead-tin yellow), it coordinates with the fatty acid chains.

Inclusion Component	Chemical Identity	Analytical Marker
Saturated Soap	Lead palmitate (C_{16}), Lead stearate (C_{18})	FTIR peak ~1540 cm^{-1} [5]
Dicarboxylate Soap	Lead azelate	ssNMR 207Pb skew/span [5]
Carbonate Phase	Lead carbonate	XRD / FTIR [10]
Mineral Phase	Synthetic plumbonacrite	Micro-XRD [6]

One of the earliest documented instances of lead soap formation is found in the thirteenth-century Westminster Retable, indicating that this is a long-term chemical process inherent to the oil medium.[10] In Francisco de Goya's *Portrait of Don Andrés del Peral* (NG 1951), the inclusions in the ground layer—which contains red lead—are large enough to protrude through the pictorial layers and be visible to the naked eye.[11] Similarly, in John Singer Sargent's *Madame X*, soap inclusions have been identified growing just beneath the upper paint layer, threatening the surface's famous smooth finish.[8]

Zinc Soaps and the Modern Crisis

Zinc oxide (zinc white, ZnO) became a staple of the artist's palette in the nineteenth century as a non-toxic alternative to lead white, but its reactivity has led to catastrophic conservation challenges.[15] Unlike lead soaps, which often form discrete protrusions, zinc soaps are frequently associated with widespread delamination, brittleness, and intralayer cleavage.[16] It is estimated that approximately 70% of paintings in museum collections are affected by some

form of metal soap-related degradation.[15]

The crystalline structure of zinc oxide can interfere with the natural oxidative cross-linking of the oil binder. Zinc oxide particles tend to entrap unsaturated free fatty acids, preventing them from connecting to the larger polymer network.[16] This makes zinc-containing paints extremely prone to "intralayer failure," where a single layer of paint splits internally.[16] This is frequently observed in Abstract Expressionist works at the Hirshhorn Museum, where mid-twentieth-century artists often used zinc-containing house paints.[9] For example, in paintings by Franz Kline and Hans Hofmann, upper layers of cadmium colors or carbon blacks have been found to crack and delaminate specifically where they were applied over a zinc-rich underlayer.[19]

Failure Mode	Visual Evidence	Chemical Cause
Intralayer Cleavage	Paint split within one layer	ZnO interference with oil cross-linking [16]
Interlayer Delamination	Paint flaking off ground	Zinc soap accumulation at interfaces [9]
Brittleness	Fine parallel cracks	High tensile stress in aged ZnO films [9]
Protrusions	"Seed-like" lumps	Aggregation of zinc stearates/oleates [9]

In the most severe cases, such as the works of the Portuguese artist Silva Porto, zinc soap formation in the ground layers caused "worm-like raised lines" and led to the total loss of the paint film, revealing the underlying support.[9] Unlike lead soaps, zinc soaps show a strong preference for accumulating at the interfaces between paint and ground, acting as a "lubricant" that facilitates peeling.[16]

Technical Analysis of Lead-Based Pigment Interactions

The reactivity of lead-containing pigments such as red lead (Pb_3O_4) and lead-tin yellow Type I (Pb_2SnO_4) is a primary driver of saponification in European oil paintings.[8] These pigments act as sources of reactive lead ions that facilitate the hydrolysis of the oil medium and subsequently react with the liberated fatty acids.

Lead-Tin Yellow Type I and Red Lead Reactivity

Lead-tin yellow Type I was the standard yellow of the Renaissance and Baroque periods. Analytical studies of paintings such as Lorenzo Costa's *A Concert* (NG 2486) have shown that the resulting soap inclusions are inhomogeneous, often possessing a lead-rich lamellar structure at the center surrounded by a more fluorescent soap matrix.[12] The inclusions commonly contain lead carbonate ($PbCO_3$), which may form as a remineralization product when lead soaps react with atmospheric carbon dioxide.[5]

Red lead, often used in grounds or as an orange-red pigment, is similarly reactive. Because lead ions act as a siccative (drying agent), red lead was frequently added to oil mordants for gilding to accelerate the polymerization of the oil.[5] However, this same siccative effect stems from the lead's ability to facilitate chemical changes in the oil, which eventually leads to the formation of disfiguring soaps.

Optical Transformations: Increased Transparency

A significant consequence of lead soap formation is the marked increase in the transparency of the paint layer.[5] As the pigment particles (such as lead white) dissolve to form soaps, the refractive index of the resulting metal carboxylate matrix ($n \approx 1.48–1.55$) becomes closer to that of the aged oil binder ($n \approx 1.50–1.60$).[2] This reduces light scattering, causing the paint to become translucent.

This phenomenon can reveal underlying "pentimenti" (artist changes), preparatory drawings, or the grain of the wood support. In Meyndert Hobbema's *Village Among Trees* (1655), the sky paint has become so transparent that the dark pores of the oak wood grain are now visible to the naked eye.[5] Such alterations can fundamentally distort the artist's original spatial and tonal intentions.

Northern European Painting Methods: 1400–1550

The materials and techniques of Northern European masters, particularly the Flemish Primitives, established the foundation for the Western oil painting tradition.[20] These artists utilized a sophisticated multi-layered approach, often combining different binders and pigments to achieve unparalleled depth and luminosity.[22]

Support, Ground, and Design

Northern European panels were typically constructed from radial-cut oak and prepared with a ground of chalk (calcium carbonate) in an animal glue medium.[22] This was followed by an "imprimatura" or isolation layer, often a thin wash of lead white in oil, which served to establish a

mid-tone and reduce the absorbency of the ground.[22]

Underdrawing was central to the design process, often executed in charcoal or a fluid medium such as iron gall ink with a brush.[22] Detailed technical studies of five paintings by Rogier van der Weyden at the National Gallery confirm a consistent use of highly detailed underdrawings that were systematically covered by thin, transparent glazes.[21]

Pigment Palette and Layering

The Northern European palette was rich and varied, incorporating both natural minerals and early synthetic pigments. Lead-tin yellow Type I provided the bright yellows, while azurite and natural ultramarine were used for blues.[22] Interestingly, research shows that egg tempera was frequently used for the initial underpainting even in works primarily executed in oil, a technique that allowed for faster drying and greater precision in the early stages of the painting.[22]

Layer Type	Dominant Materials Identified	Purpose/Function
Panel	Oak (Quercus spp.)	Rigid, stable support [22]
Ground	Chalk ($CaCO$) / Animal glue	Smooth, white surface [22]
Imprimatura	Lead white / Drying oil	Tone and suction control [22]
Underdrawing	Charcoal or Brush/Ink	Compositional layout [22]
Paint Layers	Pigments in Linseed/Walnut oil	Form, color, and texture [22]
Glazes	Red lakes / Copper greens in oil	Depth and luminosity [22]

In Gossaert's *Adoration of the Kings* and works by Wolf Huber, technical analysis has revealed the use of "prepolymerized" or heat-bodied oils, which were used to achieve the characteristic glossy and enamel-like finishes associated with the period.[21]

Anthony van Dyck: Technical Case Study of Media

Variation

The career of Anthony van Dyck provides a unique opportunity to study how an artist adapted his materials based on geography and availability. Analysis of ten Van Dyck paintings at the National Gallery, London, reveals a distinct shift in the choice of drying oils corresponding to his movements between Italy, Antwerp, and England.[27]

Geographical Shifts in Binder Choice

Van Dyck predominantly used linseed oil for his works executed in England.[27] However, during his time in Italy, Antwerp, and Brussels, he made significantly greater use of walnut oil.[27] Walnut oil was often favored for its lighter color and lower tendency to yellow over time, making it ideal for the delicate flesh tones and pale skies found in his continental portraits.

To manage the drying of certain colors, Van Dyck utilized specific additives. Softwood pitch was often added to brown glazes to provide a translucent and warm appearance, while pine resin was frequently incorporated into red lake glazes to increase their depth and saturation.[27] For pigments with poor drying properties, such as organic blacks (charcoal or bone black), he employed heat-bodied oils to ensure a stable film.[27]

Palette and Technique in "Portrait of a Woman and Child"

Technical analysis of Van Dyck's *Portrait of a Woman and Child* (NG 3011) illustrates a masterful but limited palette. Notably, for the child's purple satin dress, Van Dyck did not use the common mixture of ultramarine and red lake. Instead, he achieved the hue using a mixture of lead white, charcoal black, and a deep red lake.[25] This highlights his ability to simulate complex colors through tonal mixing rather than relying on expensive mineral pigments.

Painting Feature	Pigment Identified	Analytical Technique
Flesh Tones	Lead white, vermilion, ochres	SEM-EDX [25]
Purple Dress	Lead white, charcoal black, red lake	XRR / XRD [25]
Grey Sky	Smalt, lead white, black	SEM-EDX [25]
Green Landscape	Azurite, yellow ochre, black	XRD [25]

Despite the observations of contemporary chroniclers like de Mayerne, who suggested Van

Dyck used gum or glue media for blue and green pigments, National Gallery analyses found no evidence of such binders in his pictorial layers.[27] His technique remained essentially oil-based, utilizing the unique properties of different drying oils and resins to achieve his signature painterly effects.

Red Lake Pigments: Substrate and Dyestuff Technology

Red lakes are a class of pigments created by precipitating an organic dyestuff—such as madder root, cochineal, or kermes—onto an inorganic substrate.[26] These pigments are essential for creating deep, transparent glazes, but they are notoriously prone to fading and chemical alteration.

Chemical Complexity of the Substrate

While the substrate is often assumed to be simple hydrated alumina, research by Jo Kirby and Marika Spring has demonstrated that historical red lake substrates are chemically complex.[26] Elemental analysis often reveals significant traces of sulfur, phosphorus, silicon, magnesium, potassium, and calcium.[26] These elements originate from the raw materials used in the manufacture, including the alkali sources (lye from wood ash, lime, or chalk) and the biological sources of the dyes themselves.[29]

For example, Brazilwood lakes frequently contain high proportions of calcium, reflecting the use of calcium-containing reagents like eggshells or cuttlefish bone during precipitation.[29] The presence of phosphorus and sulfur is often linked to the proteins in the insect-derived dyes (cochineal and kermes) or the plant material in madder roots.[29]

Dyestuff Source	Key Elemental Signatures	Typical Composition
Madder Root	$S, P, Si, Ca,$	Precipitation on Alumina [29]
New World Cochineal	P, S, K, Cl, M	High dyestuff content (10-19%) [29]
Kermes (Old World)	P, Mg, C	Alum-based precipitate [29]
Brazilwood	Ca, A	Calcium-heavy substrates [29]

Technological Evolution: The Advent of Carmine

The seventeenth and eighteenth centuries saw a shift toward the production of "carmine," a highly concentrated precipitate of cochineal dyestuff (primarily carminic acid) that contains very little mineral substrate.[30] This development was reportedly a matter of chance; an accidental addition of cream of tartar (potassium hydrogen tartrate) to a dyestuff extract resulted in a pure red precipitate that was intensely vivid and transparent.[30] Carmine became the preferred pigment for miniature painting and delicate glazes, although it remained expensive due to the Spanish monopoly on Mexican cochineal.[30]

Analytical Frameworks and the Importance of Reference Materials

The study of these complex degradation phenomena relies on a rigorous analytical framework that compares samples from historical works with well-characterized standards.

The Role of Reference Collections

The Getty Conservation Institute (GCI) and other major institutions maintain extensive reference collections of pigments, drying oils, resins, waxes, and historical paint tubes.[31] These collections, such as the GCI Reference Collection (established in the early 1990s), contain over 16,000 samples, including 76 unique classes of synthetic organic pigments.[31] These standards are essential for validating the results of techniques like Py-GC-MS, which can identify synthetic binders and pigments in complex mixtures, such as those used by Jackson Pollock.[33]

Advanced Instrumentation in Conservation Science

Modern conservation laboratories employ a suite of sophisticated instruments to map chemical species at the micron scale:

- **FTIR and Micro-FTIR:** Essential for identifying the characteristic carboxylate peaks (e.g., the 1540 cm^{-1} zinc soap peak) and the presence of resins like pine or mastic.[4]
- **GC-MS and UHPLC:** Used to quantify fatty acid ratios (e.g., P/S ratios) to identify oil types and to detect specific dyestuffs like carmine in Vermeer's works.[4]
- **SEM-EDX:** Provides elemental mapping to track the depletion of potassium in smalt glass or the segregation of lead and tin in soap inclusions.[1]
- **Solid-State NMR (ssNMR):** Allows for the structural characterization of insoluble lead soaps (lead palmitate, stearate, and azelate) without requiring crystallization, a feat impossible for X-ray diffraction.[5]
- **Synchrotron-Based Techniques (XRF, XANES):** Provide high-resolution mapping of metal coordination environments, distinguishing between unreacted pigments and the resulting metal soaps in 15th-century paintings like Jan van Eyck's *The Crucifixion*.[7]

Environmental Control and Conservation Strategies

The degradation mechanisms discussed—potash leaching, saponification, and red lake fading—are all profoundly influenced by the environment. High relative humidity (RH) and temperature are the primary drivers of moisture-mediated transport of fatty acids and ions.[7]

Moisture as a Catalyst

Water acts as a catalyst for the hydrolysis of triglycerides in the oil binder, releasing the free fatty acids necessary for soap formation.[5] Furthermore, moisture facilitates the migration of these acids and metal ions through the paint film. Conservation scientists at the Met have noted that aqueous cleaning solutions, traditionally considered safe, can actually accelerate lead soap formation in some works.[7]

Mechanical Failure and Inherent Vice

As oil paint ages and undergoes saponification, it becomes increasingly brittle. Zinc white-containing paints, in particular, exhibit extreme brittleness even in relatively young films.[16] Stress/strain testing using texture analyzers has demonstrated that paints with as little as 3.6% zinc oxide crack at significantly larger bend diameters compared to lead-based or titanium-based paints.[16] This mechanical instability means that even minor environmental fluctuations can lead to catastrophic cracking and delamination.

Environmental Parameter	Impact on Paint Film	Consequent Failure
High Humidity ($>$)	Hydrolysis of oil binder	Soap formation/leaching [7]
Low Humidity ($<$)	Increased brittleness	Cracking and flaking [14]
Temp. Fluctuations	Differential layer expansion	Tenting and blind cleavage [17]
Aqueous Solutions	Facilitates ion transport	Accelerated saponification [7]

The findings presented here underscore that the "healthy" appearance of a painting can be deceptive. Chemical changes such as the leaching of potassium or the formation of sub-surface soap aggregates may be occurring long before they manifest as visible defects. The integration of technical bulletin data from the National Gallery, the Met, and the Getty provides a necessary foundation for proactive conservation. By understanding the specific

chemical triggers—such as the relationship between smalt and potash glass or the reactivity of zinc oxide—conservators can better calibrate museum environments and treatment protocols to preserve the physical integrity of these cultural artifacts. Future strategies must continue to bridge the gap between materials science and art historical context, recognizing that the artist's choice of materials dictates a unique chemical destiny for every work of art.

Citerade verk

1. Investigation of Pigment-Medium Interaction Processes in Oil Paint, hämtad mars 9, 2026, https://www.researchgate.net/profile/Catherine-Higgitt/publication/233619947_Investigation_of_Pigment-Medium_Interaction_Processes_in_Oil_Paint_Containing_Degraded_Smalt/links/56a3f2ec08aeef24c589d2e6/Investigation-of-Pigment-Medium-Interaction-Processes-in-Oil-Paint-Containing-Degraded-Smalt.pdf
2. Understanding and Preserving Smalt Pigment in Art, hämtad mars 9, 2026, https://www.naturalpigments.com/artist-materials/understanding-smalt-pigment-its-history-degradation-and-preservation-in-art
3. Analysis of Degraded Smalt Pigment | PDF | Cobalt | Paint - Scribd, hämtad mars 9, 2026, https://www.scribd.com/document/371827953/Chemistry-of-Deteriorated-Smalt-Pigment-Marika-Spring
4. Investigation of Pigment-Medium Interaction Processes in Oil Paint ..., hämtad mars 9, 2026, https://www.nationalgallery.org.uk/research/research-resources/technical-bulletin/investigation-of-pigment-medium-interaction-processes-in-oil-paint-containing-degraded-smalt
5. Investigating the Formation and Structure of Lead Soaps in ..., hämtad mars 9, 2026, https://www.metmuseum.org/essays/lead-soaps-oil-paintings
6. Full article: Lead soaps in paintings: Friends or foes?, hämtad mars 9, 2026, https://www.tandfonline.com/doi/full/10.1080/00393630.2016.1232529
7. Investigation of Heavy-Metal Soap Deterioration in Oil Paintings, hämtad mars 9, 2026, https://www.metmuseum.org/essays/heavy-metal-soap-oil-paintings
8. Elemental and Molecular Segregation in Oil Paintings due to Lead, hämtad mars 9, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5599643/
9. Condition Problems Related to Zinc Oxide Underlayers, hämtad mars 9, 2026, https://www.researchgate.net/publication/261965383_Condition_Problems_Related_to_Zinc_Oxide_Underlayers_Examination_of_Selected_Abstract_Expressionist_Paintings_from_the_Collection_of_the_Hirshhorn_Museum_and_Sculpture_Garden_Smithsonian_Institution
10. Pigment-medium Interactions in Oil Paint Films containing Red Lead ..., hämtad mars 9, 2026, https://www.nationalgallery.org.uk/research/research-resources/technical-bulletin/pigment-medium-interactions-in-oil-paint-films-containing-red-lead-or-lead-tin-yellow
11. Pigment-medium Interactions in Oil Paint Films containing Red Lead, hämtad

mars 9, 2026, https://www.nationalgallery.org.uk/media/15504/higgitt_spring_saunders2003.pdf
12. Pigment-Medium Interactions in Oil Paint Films Containing Lead, hämtad mars 9, 2026, https://cool.culturalheritage.org/waac/wn/wn27/wn27-2/wn27-204.pdf
13. Soap Formation in National Gallery Oil Painting - Molecular Vista, hämtad mars 9, 2026, https://molecularvista.com/applications/soap-formation-oil-painting/
14. There Is a Secret Villain Threatening the World's Masterpieces. It's, hämtad mars 9, 2026, https://news.artnet.com/art-world/scientists-lead-soaps-oil-paintings-936128
15. Metal Soaps in Art | springerprofessional.de, hämtad mars 9, 2026, https://www.springerprofessional.de/metal-soaps-in-art/16572460
16. Zinc Oxide – Reviewing the Research | Just Paint, hämtad mars 9, 2026, https://justpaint.org/de/zinc-oxide-reviewing-the-research/
17. Paint Consolidation - MediaWiki - AIC Wiki, hämtad mars 9, 2026, https://www.conservation-wiki.com/wiki/Paint_Consolidation
18. Metal Soaps in Art | springerprofessional.de, hämtad mars 9, 2026, https://www.springerprofessional.de/en/metal-soaps-in-art/16572460
19. CONDITION PROBLEMS RELATED TO ZINC OXIDE UNDERLAYERS, hämtad mars 9, 2026, https://repository.si.edu/bitstreams/75400625-4430-41eb-bb9a-1ca468cd1fee/download
20. National Gallery Technical Bulletin: Vol 18 - Google Books, hämtad mars 9, 2026, https://books.google.com/books/about/National_Gallery_Technical_Bulletin.html?id=23XrAAAAMAAJ
21. Technical Bulletin Vol. 18 | Research | National Gallery, London, hämtad mars 9, 2026, https://www.nationalgallery.org.uk/research/research-resources/technical-bulletin/technical-bulletin-volume-18
22. Methods and Materials of Northern European Painting in the ..., hämtad mars 9, 2026, https://www.nationalgallery.org.uk/research/research-resources/technical-bulletin/methods-and-materials-of-northern-european-painting-in-the-national-gallery-1400-1550
23. Chapter 4 - UvA-DARE (Digital Academic Repository), hämtad mars 9, 2026, https://pure.uva.nl/ws/files/2417990/148544_09.pdf
24. Uva-Dare (Digital Academic Repository) | PDF | Paintings - Scribd, hämtad mars 9, 2026, https://www.scribd.com/document/736895661/148560-25
25. Anthony van Dyck, Portrait of a Woman and Child - ColourLex, hämtad mars 9, 2026, https://colourlex.com/project/van-dyck-portrait-woman-child/
26. The Technology of Red Lake Pigment Manufacture - National Gallery, hämtad mars 9, 2026, https://www.nationalgallery.org.uk/research/research-resources/technical-bulletin/the-technology-of-red-lake-pigment-manufacture-study-of-the-dyestuff-substrate
27. Van Dyck's Paint Medium | Technical Bulletin | National Gallery ..., hämtad mars 9,

2026,
https://www.nationalgallery.org.uk/research/research-resources/technical-bulletin/van-dyck-s-paint-medium
28. Carmine in Vermeer's Palette, hämtad mars 9, 2026,
https://www.essentialvermeer.com/palette/palette_carmine.html
29. The Technology of Red Lake Pigment Manufacture - National Gallery, hämtad mars 9, 2026,
https://www.nationalgallery.org.uk/upload/pdf/kirby_spring_higgitt2005.pdf
30. The Technology of Eighteenth- and Nineteenth-Century Red Lake, hämtad mars 9, 2026,
https://www.researchgate.net/publication/233489248_The_Technology_of_Eighteenth-_and_Nineteenth-Century_Red_Lake_Pigments
31. GCI Reference Collection Development, hämtad mars 9, 2026,
https://primastoria.files.wordpress.com/2014/10/gci-reference.pdf
32. Conservation Institute - Getty Museum, hämtad mars 9, 2026,
https://www.getty.edu/conservation-institute/
33. Py-GC/MS applied to the analysis of synthetic organic pigments, hämtad mars 9, 2026,
https://scispace.com/pdf/py-gc-ms-applied-to-the-analysis-of-synthetic-organic-4yq1o6xeo5.pdf
34. Analysis of Modern Paints - Getty, hämtad mars 9, 2026,
https://www.getty.edu/conservation/publications_resources/books/analysis_modern_paints.html